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**A HYPERSONIC VEHICLE APPROACH TO
PLANETARY EXPLORATION**

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Abstract

An enhanced Mars network class mission using a lifting hypersonic entry vehicle is proposed. The basic vehicle, derived from a mature hypersonic flight system called SWERVE, offers several advantages over more conventional low L/D or ballistic entry systems. The proposed vehicle has greatly improved lateral and cross range capability (e.g., it is capable of reaching the polar regions during less than optimal mission opportunities), is not limited to surface target areas of low elevation, and is less susceptible to problems caused by Martian dust storms. Further, the integrated vehicle has attractive deployment features and allows for a much improved evolutionary path to larger vehicles with greater science capability. Analysis of the vehicle is aided by the development of a Mars Hypersonic Flight Simulator, from which flight trajectories are obtained. Atmospheric entry performance of the baseline vehicle is improved by a deceleration skirt and transpiration cooling system which significantly reduce TPS (Thermal Protection System) and flight battery mass. The use of the vehicle is also attractive in that the maturity of the flight systems make it cost-competitive with the development of a conventional low L/D entry system. Finally, the potential application of similar vehicles to other planetary missions is discussed.

Introduction

In terms of both future manned and unmanned exploration missions, Mars has continued as the object of much fascination. A variety of proposals subsequent to the Mars Observer (MO) mission are now under consideration, varying from relatively simple network missions to complex manned exploration missions. In the near term, network missions such as MESUR (Mars Environmental SURvey)¹ propose to emplace a variety of experiment packages at different locations on the Martian surface to better characterize the surface and atmosphere. These include investigation of global seismicity, internal structure, elemental chemistry of near surface soils/rocks, temperature mineralogy, fine scale structure of the surface, and atmospheric structure. In addition to advancing basic science, the unmanned missions offer an important precursor to the eventual manned missions which are expected to occur.

Prior to the relatively recent considered network missions, Mars Rover Sample Return (MRSR) Missions were investigated.^{2,3,4} Far more ambitious (and therefore expensive), the

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typical missions would include a high L/D entry system for accurate landing, a deployable roving vehicle and a means of returning the collected samples to the Earth in pristine condition. While this would indeed have greatly expanded Mars science, the developmental costs were regarded as prohibitive, particularly in the current budget environment. At the other extreme, the network class missions are limited by their greatest attribute - relative simplicity. This simplicity greatly constrains accessibility to the local terrain which might otherwise be explored by larger payloads deploying mobile instruments. Further, the ballistic or low L/D entry systems on which network missions are currently based result in, at best, a more expensive evolutionary path to missions of greater science capability. What is proposed in the current work is a means to improve the capability of potential Mars missions by use of a particular entry system and yet remain within the 'Discovery Class' (\$150 Million or less) cost ceiling.

The primary point of departure is the proposed use of an entry system adapted from existing high L/D technology. While lifting vehicles for Mars as well as other planetary missions have been seriously investigated in the past,⁵ they have typically been based on new designs requiring an expensive development effort. This in turn has driven up the overall mission costs to inordinately high levels. The existing vehicle technology that was chosen is based on the SWERVE (Sandia Winged Energetic Re-entry Vehicle) developed for the U. S. Department of Energy. The vehicle, a sharp 10.5 degree cone with short wings arranged in a cruciform (L/D > 1), has been intensely investigated and flown three times in the 2.75 m long version.

A characteristic of the vehicle is the high maneuverability at hypersonic speeds near 7 km/s at altitudes of 30-75 km (this corresponds to Mars entry velocities as well as Mars atmospheric density at altitudes of 0-60 km). This flight range made the vehicle configuration an attractive candidate as a hypersonic flight experiment platform in the proposed HYFLEX (HYpersonic FLight EXperiment) program.^{6,7} In one of the configurations, the vehicle was 6.7 m in length and required a non-ablating TPS (Thermal Protection System) in order to study the desired boundary layer transition effects. An even larger version was previously proposed as an 8 m reusable manned space plane capable of earth entry speeds greater than 7 km/s.⁸ Thus, the existence of the mature flight technology, the maneuverability at equivalent Mars density altitudes, the potential of sustaining Mars entry velocities, and finally, the similarity of entry heating in a CO₂ atmosphere, made the vehicle a potential candidate for a Mars mission.

Mission Description

A typical sequence for a Mars mission using the baseline vehicle is provided in Figure 1. The integrated vehicle, which includes the trans-Mars injection (TMI) stage, cruise stage and entry

system, is deployed in low earth orbit (LEO) via the Shuttle orbiter (launches using an expendable launch vehicle are discussed later). Here, the vehicle remains for the correct opportunity for the TMI, where a solid rocket motor provides the necessary ΔV . In the cruise phase, the transit to Mars lasts approximately 7 months during which time the communication, guidance, navigation and mid-course correction are provided by the aft-mounted cruise stage. During the Mars atmospheric entry phase, most of the cruise stage is jettisoned. A deceleration system remains on the aft end of the vehicle to decrease the ballistic coefficient, $\beta (= m/C_d A)$, thereby slowing the vehicle in the initial entry phase. Once the desired atmospheric velocity is attained, the deceleration system is released and the vehicle begins controlled flight (i.e., usage of the wing elevons and lifting flight begins). Upon achieving the desired cruise altitude, the entry vehicle is maneuvered to the desired target area. In the case of a polar or near-polar mission, the vehicle undergoes a gradual banking maneuver. Once the target area is reached, the entry vehicle is flared or stalled such that sonic velocity is achieved. At this point, the payload is jettisoned through the aft end of the entry craft, and the latter is allowed to crash. The payload package begins a gentle final descent via parachute. On landing, the various surface and atmospheric experiments are deployed.

A method to further extend the utility of small surface payloads is also shown in the Figure. This entails the deployment of a small 10 kg-class balloon system carrying a small meteorological payload and communication system. Once the characterization of the landing site is accomplished, the balloon system can be released in a desired wind direction (for example, around the polar region circumference or through a canyon system). Data would be transmitted to the payload ground station or directly to Earth as the opportunity availed itself. The device would be of relatively short duration but would provide instrument mobility beyond the immediate landing site, thereby extending the science capability of small experiment payloads.

In terms of mission planning, Mars mission opportunities occur at approximately 2.1 year intervals, with typical minimum C3 trajectories ranging in value from 8-10 km²/s². It is desirable to limit the hyperbolic approach velocity, V_∞ , to 5 km/s which results in an atmospheric entry velocity, V_E of around 7 km/s. The earliest opportunity for a simple mission would occur in late 1996, with an arrival approximately 7 months later.

Baseline Vehicle Description

A layout of the baseline 2.75 m vehicle is provided in Figure 2. The vehicle is separated into 3 main sections which include the cruise stage, the entry/flight system, and the payload section. The cruise stage contains the attitude control and course correction propellant (mid-course correction requires approximately 200 m/s), star tracker, sun sensor, and communication subsystems. During the cruise phase, the communication with the craft is accomplished via the exterior mounted, medium gain antenna. The cruise stage retains control of the craft until the final course correction at which point the entry attitude is established. The vehicle is 3-axis stabilized, with a capability of spin stabilization prior to atmospheric entry. The stage was based on a modified ALAS I (Advanced Liquid Axial Engine), though a stretched version of the larger ALAS II engine was sized for the TMI as well.

The entry system, the major system element, consists of the aerodynamic vehicle used during the atmospheric entry and flight phase. The ballistic coefficient of the flight vehicle itself is 15,250 kg/m² (the decelerator system used during initial entry will decrease this by a factor of 10). The structure is comprised of machined aluminum sections over which the TPS, a silica phenolic material, is attached. In the baseline vehicle, the nose tip, wing leading edge and control surfaces, are made of a carbon-carbon composite (for heating calculations, the nose radius was assumed to be .025 m, and the wing leading edge radius, .04 m). The forward section contains the ballast, transpiration cooling system and forward flight battery compartment. The batteries were assumed to be high power density, non-rechargeable lithium thionyl chloride batteries. Flight control is accomplished with the current generation flight computer, the SANDAC V (SANDia Aerospace Computer). The navigation system is the mini-RIMS (Roll stabilized Inertial Maneuvering System), an inertial navigation system used in concert with the flight computer. For higher altitude trajectories, navigation is accomplished with accelerometer data. For low altitude flight over more perilous terrain, a radar altimeter, SITAN (Sandia Inertial Terrain Aided Navigation), may be utilized (the necessary, accurate terrain data is expected to become available from the Mars Observer mission). This may be located in one of the aft compartments and flown on the leeward side of the craft during the peak heating phase of the entry to reduce window heating. Control of the four wing elevons is accomplished with four electro-mechanical actuators located at the aft end of the payload compartment.

The payload section consists of a cylindrical canister 1.12 m long and .25 m diameter with two lobes which extend to a width of .45 m. This shape, shown in relief in Figure 2, was dictated by the requirement of rear jettison, which is somewhat impeded by the four elevon actuators (as a note, it is possible not to jettison the payload canister and land the entire vehicle. However, the size of the parachute is greatly increased and there is a problem regarding thermal soak after landing). By grouping the elevons as depicted, the total payload for the baseline vehicle is .085 m³. The payload canister carries the basic science payload, surface power and communication systems, and the descent parachute system.

The mass summary for the baseline 2.75 m vehicle is provided in Table 1. The payload system mass is 50 kg and the total vehicle mass including the cruise stage is 233 kg. The total vehicle mass is sensitive to the TPS mass, which may be tailored to the desired flight trajectory. At 2.5 cm thickness, the TPS mass is 93 kg, which represents a major fraction of the 'dry' entry mass. The battery mass is also heavily dependent on the trajectory. The payload instruments represent a network class mission and were obtained from the MESUR work.¹ As can be seen from the Table, most of the new design effort is the development of the payload, and to a lesser extent, the cruise stage.

Mars Hypersonic Flight Simulator

The difficulties associated with a lifting, maneuvering entry is that the multitude of possible entry and flight trajectories can significantly affect the vehicle design (this is the reverse of a low L/D entry in which the entry profile is heavily constrained and easier to analyze). To aid in the vehicle and mission analysis, a Mars

Hypersonic Flight Simulator was developed which allows real time flight profiles to be readily obtained. The flight simulator is a modification of one developed originally for the High Performance Space Plane⁷ and allows for either manual or pre-programmed control. Physically, the simulator resembles a cockpit and the user literally flies the vehicle through simulated Mars conditions. The simulator includes a Mars atmospheric density profile as well as terrain features shown on the map in Figure 3 (for scale, a map of the United States, approximately 4800 km across, is shown). The terrain features were found to be particularly useful in selecting flight paths based on terrain avoidance. The simulator can be readily flown between the 25 km high equatorial volcanoes which are problematic for entry systems. Ultimately, the simulator information can be used to program the desired flight trajectories into the SANDAC V flight computer.

Initially, the simulator was used for the determination of the effectiveness of aerocapture, given the initial difficulty due to the high value of the ballistic coefficient, β , of the original vehicle. It was also used to determine the sensitivity of entry velocity, V_E , entry angle, γ , and traverse distance. The simulator was also useful in estimating the TPS and battery requirements, which are significant mass components of the entry system, and in turn effect β . The battery size is estimated from the power requirements obtained from the elevon deflection profile as a function of time. The TPS requirement for the nose tip, wing leading edge, and vehicle body may be obtained directly from the simulator. For improved calculation, the atmospheric density and flight velocity profiles may be used in an improved simulation code which better corrects for CO₂ effects.

Flight Trajectories and Heating Profiles

Two different types of entry and flight trajectories are given in Figure 4. These fairly different cases illustrate the possible range in flight trajectories. For the first trajectory, case a, the vehicle undergoes atmospheric entry and attains level flight at an altitude of 20 km for approximately one hour. The ground track in this case is approximately half of the planet circumference. In case b, the vehicle flies at 5 km above the surface and traverses 4500 km in a period of 10 minutes. The resulting ground track is roughly the width of the U.S. map, imposed on the Mars Hypersonic Flight Simulator map in Figure 3. Due to the proximity of the ground, the trajectory would have to be carefully planned and the previously described guidance/navigation scheme would be desirable. In the latter case, the heating is more severe, but of significantly shorter duration. In the former case, the problem relating to the total integrated heat load and the warming of critical flight systems becomes a consideration.

The centerline heating distribution for a typical 'severe' case is provided in Figure 5. The cold wall heating profile on the top, case a, is obtained from a velocity $V = 5.2$ km/s, an altitude of $h = 5$ km, and at 0 angle of attack. The heating distribution is significantly decreased in case b, in which a decelerator has been used to reduce the velocity to 3.2 km/s at the same altitude. In the latter case, the corresponding equilibrium cold wall heating rates for a .025 m radius nose drops to 1.45 Kw/cm². The peak heating values for this latter case are within the capability of the silica phenolic and carbon carbon TPS used on the original system. However, this may be further reduced by a factor of 10 by the use of a transpiration cooling system,

which further reduces the thickness and therefore the total TPS mass fraction. A general trend is that, due to the $V^{3.2}$ term in the convective heating rate, the decelerator system is extremely useful, and made more so by entering at a shallower entry angle and allowing more time for use prior to level flight.

Another advantage in selecting the particular vehicle configuration is the existence of computational grids which calculate real gas effects for Earth atmospheric flight. These codes can be modified to include CO₂ chemistry and applied to the Mars entry problem (this effort is already underway). This information could be used in refining the heating calculations as well as updating the aerodynamic coefficients in the flight computer for improved vehicle performance. A more detailed analysis will be forthcoming.

Vehicle Launch and Scaling Issues

The baseline STS launch configuration is shown in greater detail in Figure 7. The 2.75 m vehicle with cruise stage and solid propellant motor is shown stowed in one of three vertical 'launch tubes' at the aft end of the cargo bay. The dimensions of such a storage tube are approximately .7 m wide by 4.8 m in height. The Mars vehicle would be deployed via a spring mechanism. The mass of the Mars vehicle, including solid rocket motor, is approximately 680 kg. The foregoing illustrates the ease (given STS payload and c.g. constraints) with which such a relatively small system might be accommodated, making it more attractive for 'piggy-back' opportunities.

In Figure 8, two other launch configurations are shown using an expendable launch vehicle. In the first Figure, 8a, a single vehicle is launched on top of the ELV in a shroudless configuration. Unlike other lifting vehicles, the advantage of the cone shape is that the axisymmetry results in minimum impact to the ascent flight dynamics (this continues to be a problem in current programs such as Hermes). In the second Figure, 8b, accommodating an elongated vehicle in a shorter shroud is made possible by hinging the forward section of the vehicle.

Another attribute of the SWERVE based vehicle is the inherent scalability. After the 1.75 m point (at the payload compartment forward interface), the vehicle can easily scale in length with minimum impact on the other flight subsystems. Due in part to the manner in which the vehicle is assembled, this results in only a modest increase in cost of larger vehicles. The payload cavity volume increase as a function of vehicle length is given in Figure 9. The addition of 1 m in vehicle length increases the payload cavity volume three fold from .15 to .45 m³.

As an illustration of the increased science capability which this allows, Figure 10 depicts a 3.65 m vehicle shown with a 1.3 m long semi-autonomous mini-rover with articulated joints. In maintaining the same ballistic coefficient, the payload mass for this larger vehicle is increased by over a factor of 3. At the extreme, the previously mentioned 8 m long High Performance Space Plane is shown in Figure 10 (the wings were omitted from the Figure). Features of this proposed 2800 kg vehicle included a hinged 4 m nose section to better fit within the STS cargo bay. The aft end of this size vehicle is 1.5 m in diameter, making it possible to stow a large rover of the class originally intended for MRSR missions.

Comparison of Entry Vehicle Systems

A comparison of the low L/D MESUR entry system and the high L/D system currently under discussion is in order, particularly since the two represent opposing views in design approach. The advantages of the former include the design simplicity (i.e., there are no complex subsystems required for entry), good payload fraction, low ballistic coefficient (30 kg/m^2), and the advantage of previous use in the Martian environment. However, there are disadvantages. First, the dimension of the 70° conical aeroshell is ultimately limited by the shroud diameter. To exceed this, and therefore to grow or evolve the size of the payload, a complicated deployable mechanism would be required (or, a significantly more expensive Earth launch system employed). Secondly, access to the Mars surface is constrained to lower elevations (due to sonic requirements for parachute deployment) and lower latitudes during poorer launch opportunities. In addition, there are constraints imposed due to lighting conditions during landing for descent imaging. Advantages of a SWERVE based vehicle would include greatly expanded surface coverage (including access to the polar regions during poorer launch opportunities), a superior evolutionary path to larger payloads and missions, and a greater flexibility in launch and shroud options. Other considerations would include a larger entry corridor, reduced deceleration loads, and minimum effect on the vehicle TPS from dust storms. In addition, the use of a high L/D vehicle would develop experience in flying a controlled vehicle in an alien environment prior to manned missions.

Ultimately, however, the discussion reduces to one of cost and the resulting effect on the mission. Because of the existing tooling and subsystems, the baseline SWERVE derived vehicle can be prepared for \$10-15 million, which is the approximate development cost of the simpler descent system for MESUR.⁹ Another figure of merit might include cost/kg of delivered payload. When the vagaries associated with launch costs as well as the greater payload mass resulting from a modest increase in the length of the SWERVE based vehicle are considered, this figure also may be comparable. Viewed from a longer range perspective in which larger payloads to Mars are desired, the less expensive evolutionary path afforded by the SWERVE based vehicle is attractive.

Applicability to Other Planetary Missions

The use of high L/D vehicles have been previously considered for other planetary bodies. In the case of Venus and Titan, where typical entry velocities are 12 and 13 km/s respectively, the resulting heating is appreciably higher than what has previously been considered for Mars missions. Because of a near cubic relationship between velocity and convective heating, the heating in the Venusian atmosphere is 5.6 times greater at the same density (and approximately 7.2 times for Titan). In order to maintain the same heating rate as the Mars vehicle, the vehicle must fly at a higher altitude (i.e., approximately 1/13 lower density, which, in the thick atmospheres of both planets does not result in a significantly higher altitude). In order to reduce the resulting elongation in flight time, the effectiveness of the decelerator system would increase to achieve the same drag. This becomes important as the total integrated heat load is proportional to b (as is the TPS mass for systems in which transpiration cooling is not employed), and it is necessary shed the excess kinetic energy as early as possible. In addition, transpiration cooling is more attractive, and

skipping maneuvers can also be contemplated. The attraction regarding Venus missions include the low ΔV , the frequency of opportunity, the lack of requirements regarding planetary protection, and a CO_2 atmosphere similar to Mars. In the case of Titan missions, 6.3 m biconic entry vehicles with a β of 800 kg/m^2 have been previously studied in some detail.⁸ This is of a scale possible for the elongated SWERVE derived vehicle.

Summary

A Mars mission is proposed which bridges the mission capability of network class and more capable class rover missions. The principal feature is the use of a SWERVE derived atmospheric entry system which offers certain advantages over the conventional, low L/D aeroshell. These include greater surface coverage, greater launch flexibility, and a more attractive evolutionary path to larger vehicles and payloads. The entry system is shown to perform better with the use of a decelerator package as well as a transpiration cooling system that minimizes TPS and flight battery mass. Although more complex, the existing SWERVE derived vehicle may be modified at the same approximate cost as the development of a new, though simpler, low L/D aeroshell. The basic shape may be attractive for a variety of missions for Mars and perhaps other planetary bodies.

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- ⁹Communication with Sandia Laboratory personnel.

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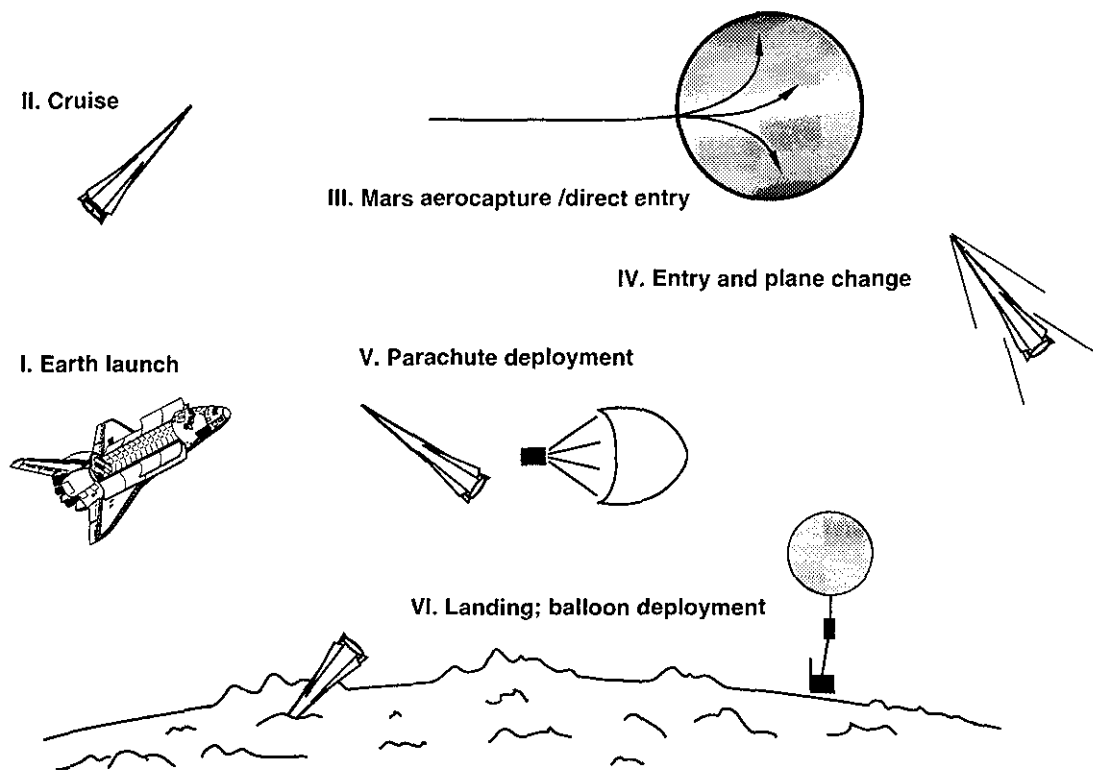


Figure 1. Mars mission description.

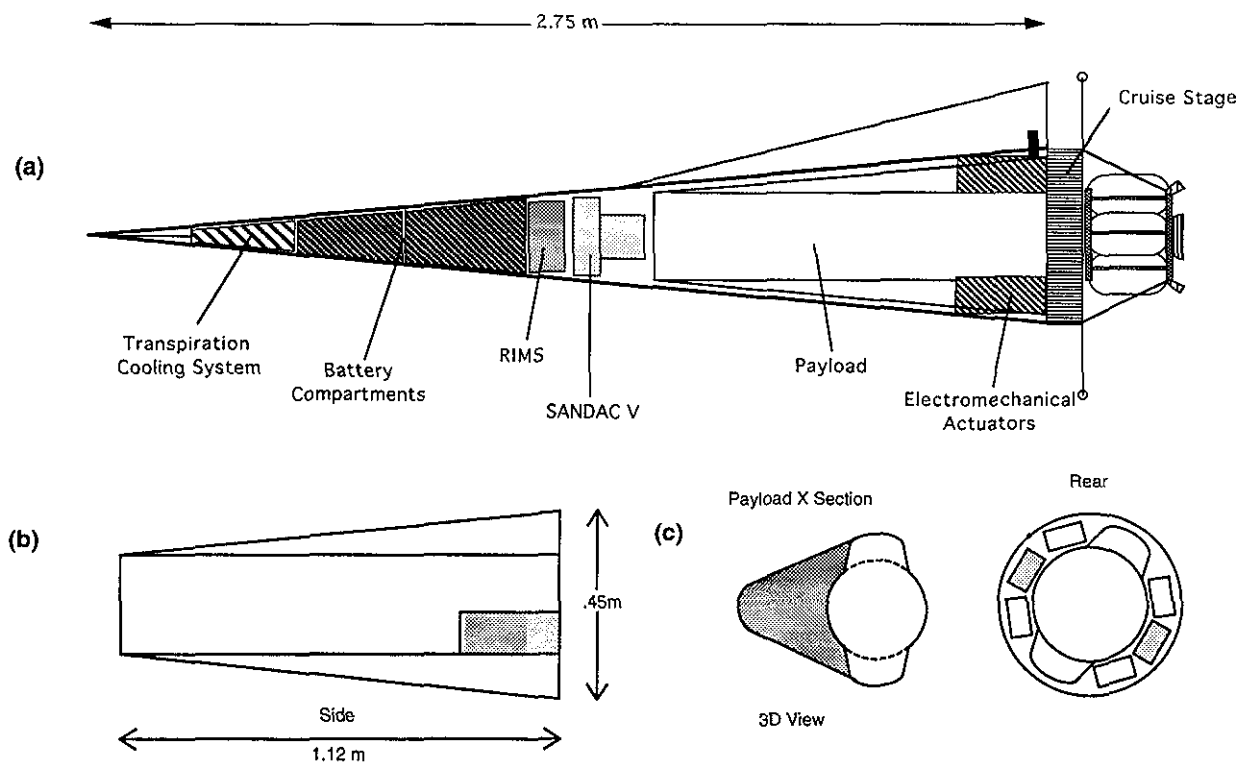


Figure 2. Baseline 2.75 m vehicle: a) vehicle layout, b) payload geometry, c) aft cross section for payload jettison.

SYSTEM	ELEMENT	MASS (KG)	MASS (KG) SUBTOTAL	HERITAGE (Costing)	COMPLEXITY (Costing)
ENTRY SYSTEM					
	Mini-rims	4.75		Off the shelf	
	Sandac V	2.75		Off the shelf	
	Actuator	17.50		Off the shelf	
	Structure	17.00		Off the shelf	
	TPS Body	90.00		Off the shelf	
	TPS- Cone	2.00		Off the shelf	
	Flight Batteries	25.00	159.00	Off the shelf	
PAYLOAD SYSTEM					
	Science				
	D & S Imager	1.84		New Design	High
	Atm Structure	1.50		Major Mod	High
	Meteorology	0.66		New Design	High
	Elemental Comp	0.60		New Design	Very High
	3-Axis Seismometer	1.50		New Design	High
	Thermal Analyzer	2.00		New Design	Very High
	Integration HW	2.00		Off the shelf	
	Structure	5.00		New Design	Medium
	Power	18.00		Mod design	Medium/low
	Harness	2.75		New Design	Low
	Communication	5.50		New Design	Medium
	Parachute	8.00		Mod Design	Medium
			49.35		
CRUISE SYSTEM					
	Propulsion (ALAS)	17.00		Mod Design	Medium
	Structure	1.00		New Design	Medium
	Pyro (Control)	1.00		Off the shelf	
	Star Tracker	0.75		New Design	Medium
	Sun Sensor	0.23		Off the shelf	
	Electronics	4.00		New Design	Low
	Antenna	1.00	24.68	Minor Mod	
Entry System Mass			208.35		
Total Mass			233.00		

Table 1. Mass summary.



Figure 3. Mars Hypersonic Flight Simulator terrain map.

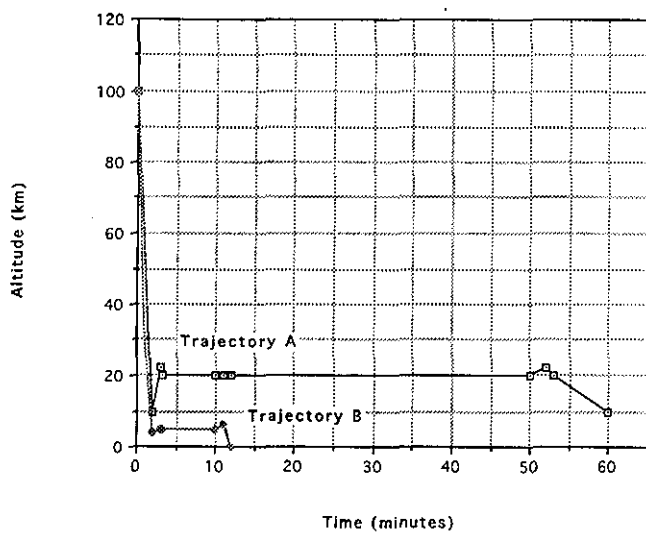


Figure 4. Mars entry trajectories: a) long traverse, (60 minutes), b) short traverse (10 minutes).

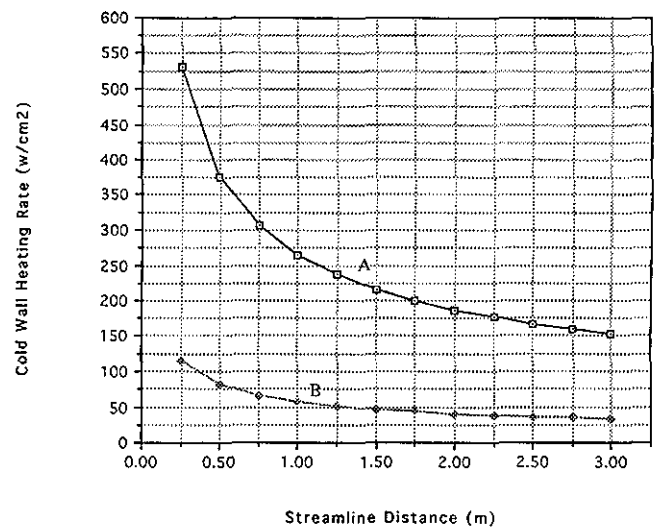


Figure 5. Streamline heating rates for two vehicle conditions.

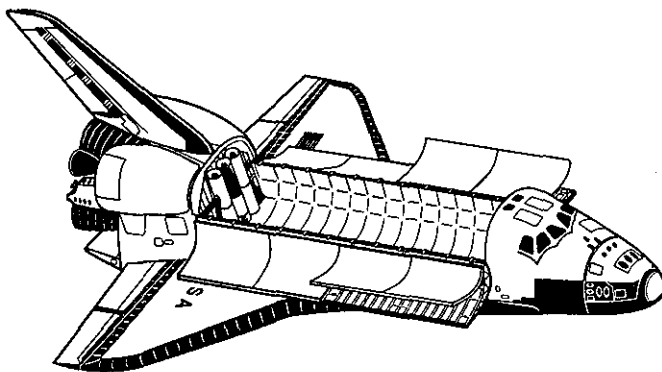


Figure 6. STS deployment.

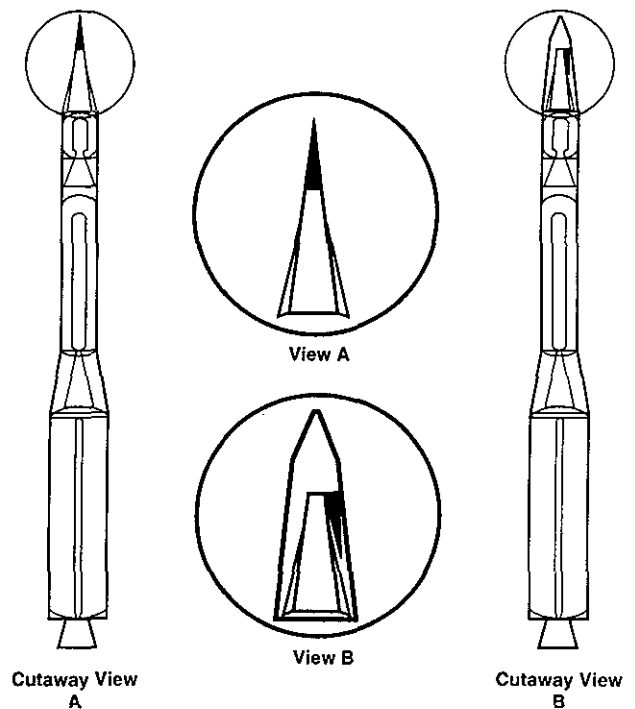


Figure 7. Ground launched vehicle deployment: a) vehicle launched without shroud, b) vehicle within constrained shroud with deployable forebody.

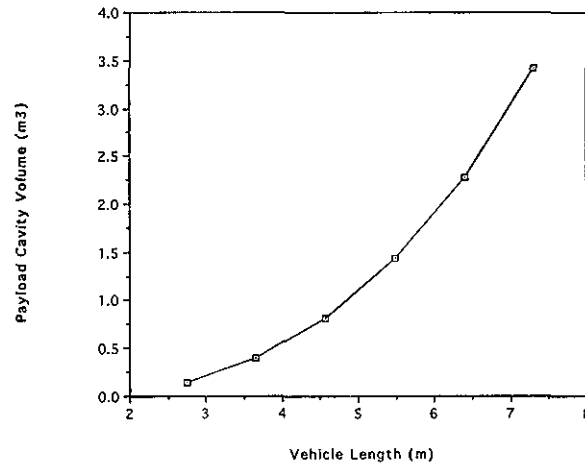


Figure 8. Payload cavity volume increase as function of vehicle length.

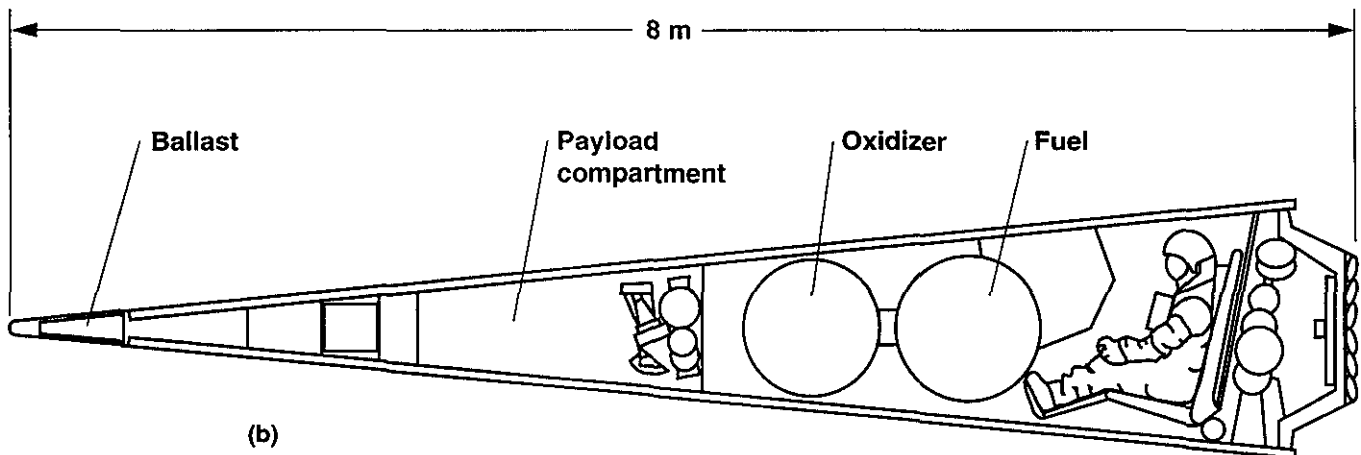
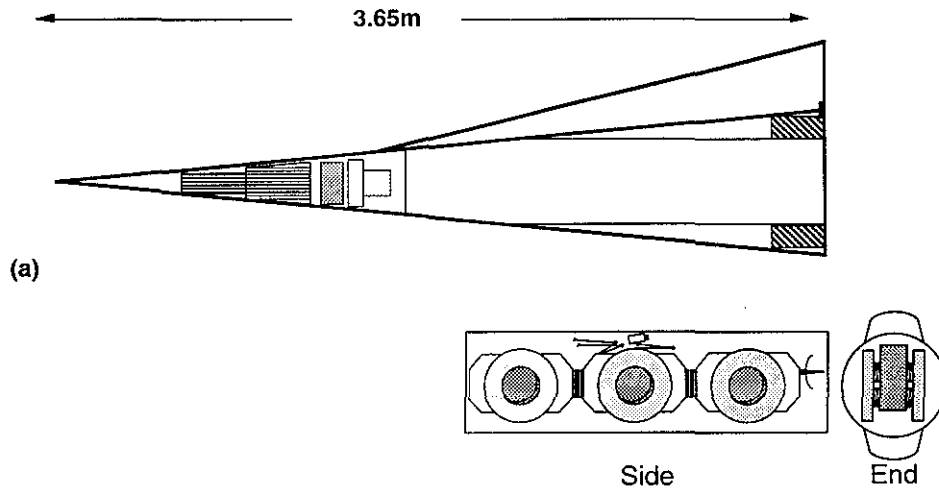


Figure 9. Examples of stretched vehicles: a) 3.65 m Mars vehicle with mini-rover, b) previously proposed 7.9 m manned vehicle for earth re-entry.